UNIVERSITY OF PERADENIYA
Faculty of Engineering
Department of Electrical & Electronic Engineering

Design & Implementation of an FM Pocket Radio

From Breadboard to Custom PCB

EE2050 • PRODUCT DEVELOPMENT

Members

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Abstract

This report documents the design, iteration, and implementation of a portable FM pocket radio, developed across three progressive iterations for the **EE2050 Product Development** module during Semester 3 (February to March 2026). The project evolved through three iterations:

- **Iteration 1:** A basic Arduino UNO + RDA5807M + TDA2822M perfboard prototype.
- **Iteration 2:** An ESP32-based feature-rich build with OLED display and rotary encoders.
- **Iteration 3:** A custom PCB design featuring an ATmega328P, PAM8403 Class-D amplifier, CH340C USB-UART bridge, and single USB-C port for both charging and programming.

The design process was heavily AI-assisted, with multiple language models consulted for circuit design, debugging, and verification. This report captures the complete engineering journey: the problems encountered, the solutions discovered, and the lessons learned.

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1 Introduction

The EE2050 module requires each group to design and build a product-grade electronic device. This group chose to build a **pocket FM radio** - a device that would evolve from a basic functional prototype to a professional PCB-based product.

1.1 Problem Statement

Design and build a pocket-size FM radio that:

- Receives FM broadcast (87.5–108.0 MHz, Sri Lanka band)
- Provides clear audio output through both speaker and headphones
- Is rechargeable via USB-C
- Fits in a pocket ($\sim$70 $\times$ 50 mm PCB)
- Can be programmed/updated via the same USB-C port
- Has a user-friendly OLED display with station names
- Costs under Rs. 5 000

1.2 Project Objectives

1. **Primary:** Build a working FM radio on perfboard (proof of concept)
2. **Secondary:** Add digital UI, battery power, and encoder controls (feature expansion)
3. **Tertiary:** Design a custom PCB with integrated charging, programming, and product-grade finish
4. **Documentation:** Maintain a complete design log capturing every decision and iteration

1.3 Key Decisions

Decision	Choice	Reason
MCU	`ATmega328P-PU` (DIP-28)	Same as Arduino UNO, 3.3 V / 8 MHz, through-hole
Radio IC	`RDA5807M` on RRD-102 V2.0	DSP demodulation, auto-seek, I^2C, already owned
Amplifier	`PAM8403` Class-D (SOP-16)	90 % efficient, $\overline{\text{SHDN}}$ mute pin, SMD
Display	`SSD1306` 0.96" OLED	I^2C, 128 $\times$ 64, already owned
Power	3.7 V LiPo + `HT7333` LDO	Low dropout ($<$100 mV), pocket-size
USB	Single USB-C port	Charging (`TP4056`) + programming (`CH340C`)
Audio	Mono speaker + stereo headphones	Single speaker sufficient for pocket radio

Table 1: Key design decisions and rationale.

1.4 Design Methodology

1. **Research phase** - component selection, datasheet review, tutorial search

2. **Prototype 1** - basic perfboard build; verify FM reception and audio
3. **Prototype 2** - add display, encoders, battery; feature validation
4. **PCB design** - custom 2-layer board in KiCad 9.0
5. **Documentation** - comprehensive build guide + this report

AI-Assisted Design

Multiple AI models (ChatGPT, Gemini, Claude, Grok, DeepSeek) were consulted throughout the project for circuit design, debugging, code generation, and design verification. This report documents both the benefits and pitfalls of this approach.

2 Design Evolution

The FM Pocket Radio project evolved through three distinct iterations, each building on lessons learned from the previous build:

- **Iteration 1:** Proof of Concept (Perfboard)
- **Iteration 2:** Feature Expansion (ESP32 + OLED)
- **Iteration 3:** Custom PCB Design

The timeline below summarises the progression from a proof-of-concept perfboard to a custom PCB design.

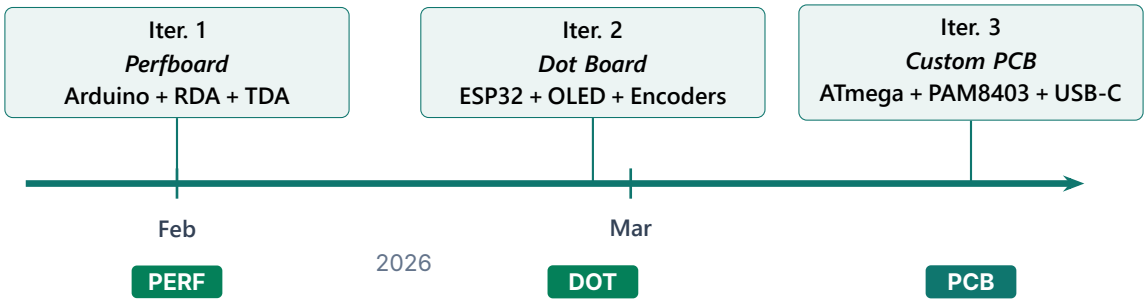


Figure 1: Design iteration timeline: February to March 2026.

2.1 Iteration 1 - Proof of Concept (Perfboard)

Period: February 2026.

- `Arduino UNO` (5 V logic, ATmega328P)
- `RDA5807M` FM receiver (RRD-102 V2.0 module)
- `TDA2822M` dual audio amplifier (DIP-8, single-ended mode)
- 4 Ω / 5 W speaker
- B100K linear volume potentiometer
- 2× tactile seek buttons (**SEEK_UP**, **SEEK_DOWN**)

LOG ENTRY 1 • Feb 17 • First build attempt

Components purchased from Duino Electronics (Rs. 230 for RDA5807M module). Built on 15×18 cm dot board. First power-on: no sound. I²C scanner found nothing at `0x10`. Switched to address `0x11` (index mode) - module responded. Then seek always stuck at 108.9 MHz. Discovered solder ball on 32.768 kHz crystal cylinder preventing PLL lock.

LOG ENTRY 2 • Late Feb • Amplifier overheating

`TDA2822M` getting extremely hot. 5 V rail collapsing from 5 V to 3 V when speaker connected. Root cause: 4 Ω speaker at max volume exceeds USB current capacity. Fix: lower software volume to 12/15, add 2.2 Ω series resistor with speaker.

Caution

Wrong IC Purchased: We accidentally bought `TEA5767` (RRD-102BC, Rs. 550) thinking it was the same as `RDA5807M` (RRD-102V2.0, Rs. 230). Same module form factor, completely different I²C protocol. Extended debugging before the mistake was identified.

Working FM radio with seek buttons and hardware volume. Multiple revisions (Rev 1.0 to 3.2) of the build document were produced as issues were discovered and resolved.

2.2 Iteration 2 - Feature Expansion (ESP32 + OLED)

Period: Late February – March 2026.

- `ESP32` 30-pin DevKit (dual-core, 240 MHz)
- `RDA5807M` FM receiver
- `TDA2822M` amplifier (BTL mode)
- `SH1106` 1.3" OLED display (128×64, I²C)
- 2× S141 rotary encoders (tuning + volume)
- 18650 Li-ion cell + `TP4056` charger module
- 15×9 cm dot board

- BTL amplifier mode (≈4× power vs. single-ended)
- Software volume via `RDA5807M` register (no hardware pot)
- OLED display: frequency, station name, signal bars, battery %, volume
- 68-station Sri Lankan FM database in firmware
- Rotary encoders for tuning and volume
- Battery power with voltage monitoring (100 k/100 k divider on `GPIO34`)

Decision

BTL Mode: Bridge-Tied Load configuration drives the speaker from both `TDA2822M` outputs (`pin 1` and `pin 3`). Neither speaker terminal connects to GND. This delivers approximately 4× the power of single-ended mode, but grounding either speaker wire will destroy the amplifier.

LOG ENTRY 3 • Late Feb • OLED UI development

Full UI built with U8g2 library: `logisoso22` font for frequency, `profont12` for labels. Signal bars + percentage, speaker icon with volume waves, battery icon with %, 20-bar dancing EQ animation. Python Colab simulation used for rapid UI iteration without reflashing ESP32.

Feature-complete prototype with premium UX. Validated all features before PCB design.

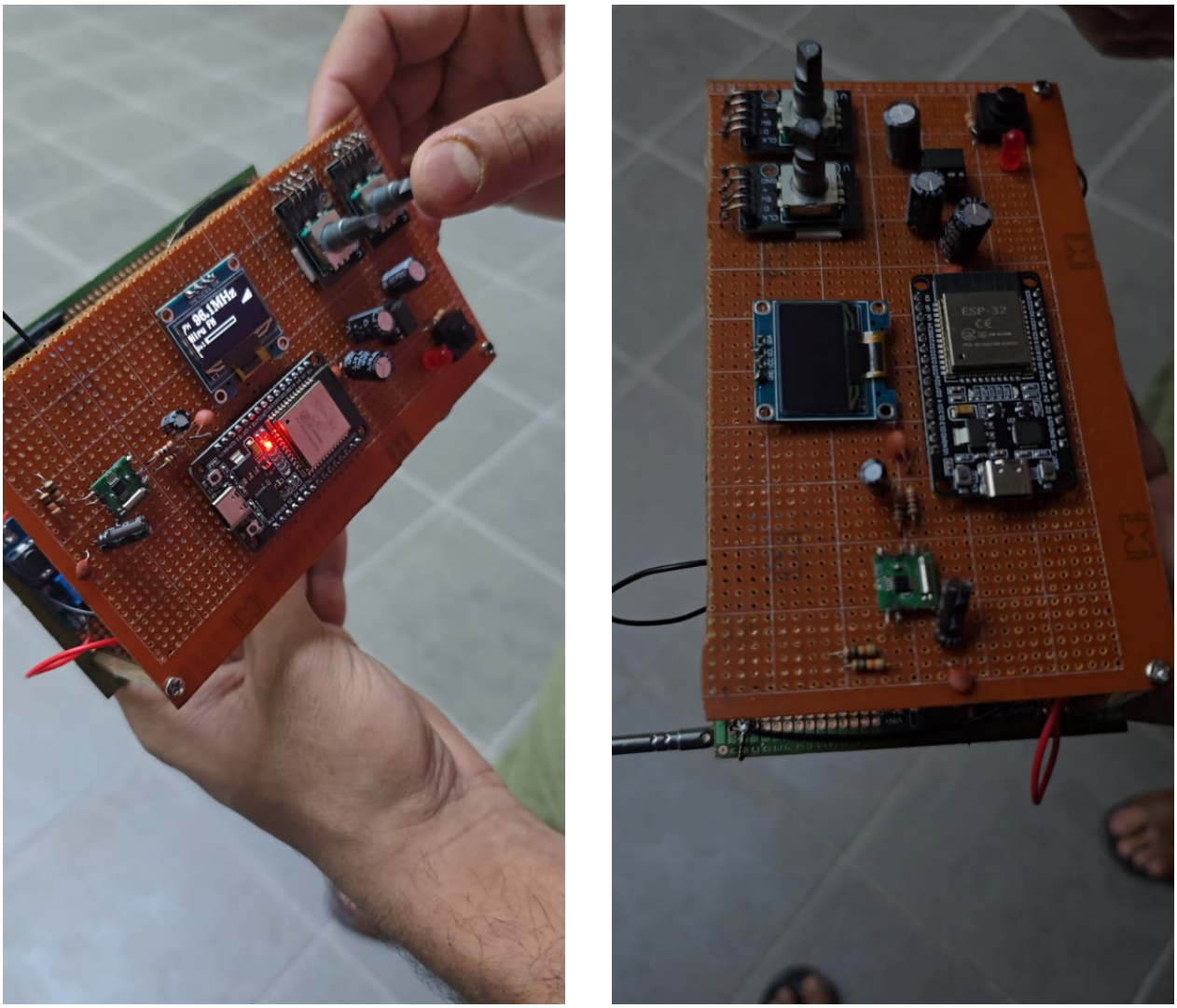


Figure 2: Iteration 2 (Dot Board) physical prototype in operation.

2.3 Iteration 3 - Custom PCB Design

Period: March 2026.

- `ATmega328P` (8 MHz internal oscillator, 3.3 V)
- `RDA5807M` FM receiver
- `PAM8403` Class-D stereo amplifier (3 W×2)

- CH340C USB-to-UART bridge
- SSD1306 0.96" OLED display (128×64)
- TP4056 linear charger + DW01A/FS8205A battery protection
- HT7333 3.3 V LDO regulator
- USB-C connector (single port: charging + programming)
- Custom 70×50 mm 2-layer PCB (designed in KiCad 9.0)

- ATmega328P replaces ESP32 (simpler, cheaper, sufficient)
- PAM8403 Class-D replaces TDA2822M (90 % efficient, no heatsink)
- Single USB-C for both charging and programming
- On-board battery protection (DW01A + FS8205A)
- Custom PCB designed in KiCad 9.0

Decision

ATmega328P over ESP32: For the final product, ESP32's WiFi/BT capabilities were unnecessary overhead. ATmega328P at 3.3 V / 8 MHz provides sufficient I/O for I^2C + OLED + encoders + ADC, runs the same Arduino ecosystem, and eliminates level-shifting complexity.

Design Status

Ready for Fabrication: The Phase 3 design exists in software (KiCad schematic + PCB layout + comprehensive build guide). It represents the *target architecture* and provides a complete foundation ready for future fabrication and assembly.

2.4 Iteration Comparison Table

Table 2: Comparison of the three design iterations.

Feature	Iter. 1	Iter. 2	Iter. 3
MCU	Arduino UNO	ESP32	ATmega328P
Radio	RDA5807M	RDA5807M	RDA5807M
Amplifier	TDA2822M (SE)	TDA2822M (BTL)	PAM8403 (Class-D)
Display	None	SH1106 1.3"	SSD1306 0.96"
Volume	Hardware pot	Software (0–15)	Software (0–15)
Tuning	Seek buttons	Rotary encoder	Rotary encoder
Power	USB 5 V	18650 + TP4056	LiPo + TP4056 + DW01A
Audio output	Speaker only	Speaker only	Speaker + headphones
Board	15×18 cm perfboard	15×9 cm dot board	70×50 mm custom PCB
Status	Built & working	Built & working	Design completed

3 System Architecture (Iteration 3)

This section describes the complete electrical architecture of the Phase 3 custom PCB design. Every subsystem - power, digital, RF, audio, and USB - is integrated onto a single 2-layer FR-4 board (∼70 × 50 mm).

3.1 Block Diagram

Figure 3 shows the top-level signal and power flow. Colour coding groups components by subsystem: power (copper), digital/USB (teal), RF (green), and audio (slate).

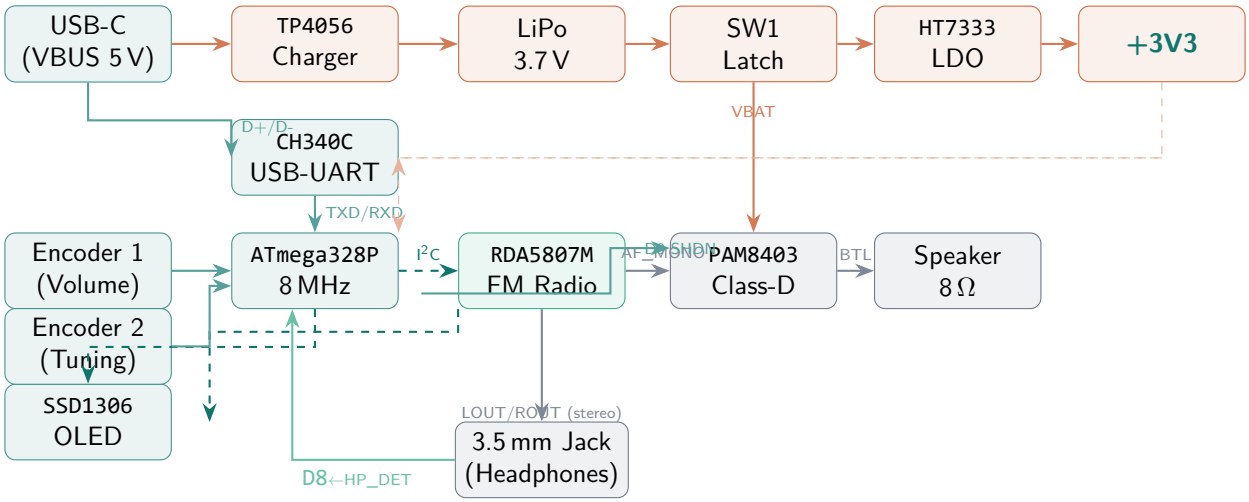


Figure 3: Phase 3 system block diagram. Dashed lines = I²C bus; solid arrows = power/signal flow.

3.2 Power Architecture

The power system is designed around a single-cell 3.7 V LiPo (800–1000 mAh) with on-board charging, protection, and regulation.

3.2.1 Charging

USB-C VBUS (5 V) feeds the TP4056 linear charger. The charge current is set by PROG resistor R9:

$$I_{\text{charge}} = \frac{1000\,\text{V}}{R_9} = \frac{1000}{2200} \approx 455\,\text{mA}$$

The TP4056 implements CC/CV charging: constant current at 455 mA until the cell reaches 4.2 V, then constant voltage until current tapers to ∼45 mA (1/10 termination).

3.2.2 Battery Protection

The LiPo BAT- terminal connects through a DW01A + FS8205A low-side protection circuit. Two back-to-back N-channel MOSFETs (FS8205A) sit in the ground return path:

- **Over-discharge:** DW01A opens the discharge FET when $V_{\text{cell}} < 2.4\,\text{V}$
- **Over-charge:** opens the charge FET above 4.3 V (backup to TP4056)
- **Short-circuit:** opens within ∼10 ms if current exceeds ∼3 A

3.2.3 Regulation

VBAT (3.0–4.2 V) passes through latching switch SW1, then enters the HT7333 LDO which outputs a stable **+3V3** rail. This rail powers:

- ATmega328P (MCU)
- RDA5807M (FM radio)
- SSD1306 OLED module
- 2× rotary encoders (pull-ups)
- CH340C (USB-UART bridge)

3.2.4 Audio Power

The PAM8403 amplifier is powered *directly* from VBAT (post-switch), bypassing the LDO. This maximises available audio power:

$$P_{\text{spk}} = \frac{V_{\text{BAT}}^2}{2 \cdot R_L} = \frac{4.2^2}{2 \times 8} \approx 1.1\,\text{W (peak)}, \quad \text{typical} \approx 0.5\,\text{W}$$

Running the amplifier from the unregulated rail also avoids injecting Class-D switching noise into the sensitive 3.3 V digital/RF supply.

3.2.5 Battery Sensing

VBAT is divided by R7/R8 (100 kΩ / 100 kΩ) and filtered by C19 (100 nF) before reaching A0 on the ATmega328P:

$$V_{A0} = V_{\text{BAT}} \times \frac{100\text{k}}{100\text{k} + 100\text{k}} = \frac{V_{\text{BAT}}}{2}$$

At full charge (4.2 V), $V_{A0} = 2.1$ V; at cutoff (3.0 V), $V_{A0} = 1.5$ V. The 100 nF capacitor compensates for the high source impedance (50 kΩ Thévenin) which exceeds the ATmega's recommended $\leq$10 kΩ for accurate ADC sampling.

HT7333 vs AMS1117 - Why Dropout Matters

At battery minimum (3.0 V), the HT7333 outputs ~2.9 V (dropout <100 mV). The ATmega328P runs reliably at 2.9 V / 8 MHz - its minimum operating voltage is 2.4 V at this frequency. The commonly-used AMS1117-3.3 requires $\geq$1 V headroom and would fail below 4.3 V battery - completely unsuitable for a single-cell LiPo design.

3.3 Audio Signal Path

3.3.1 Mono Speaker Path

The RDA5807M provides stereo line-level outputs (LOUT, ROUT). These are passively mixed to mono:

$$\text{LOUT} \xrightarrow{\text{R3 (10 k}\Omega\text{)}} \textbf{AF_MONO} \xleftarrow{\text{R4 (10 k}\Omega\text{)}} \text{ROUT}$$

The mixed signal passes through C5 (1 μF DC-blocking capacitor) into PAM8403 INL. The amplifier drives the speaker in Bridge-Tied Load (BTL) configuration via OUTL+ and OUTL-, delivering up to ~0.5 W into 8 Ω.

3.3.2 Stereo Headphone Path

For headphone output, the RDA5807M outputs are AC-coupled directly to the 3.5 mm jack:

- LOUT → C6 (10 μF) → Jack TIP (left)
- ROUT → C7 (10 μF) → Jack RING (right)

The 10 μF coupling capacitors set a -3 dB low-frequency corner of:

$$f_c = \frac{1}{2\pi \cdot C \cdot R_{\text{load}}} = \frac{1}{2\pi \times 10\,\mu\text{F} \times 32\,\Omega} \approx 500\,\text{Hz (acceptable for FM audio)}$$

3.3.3 Antenna Coupling

The headphone cable doubles as the FM antenna. A 100 pF capacitor (C8) AC-couples the Jack TIP to the RDA5807M ANT pin:

- C8 passes RF ($>$87 MHz) while blocking audio frequencies
- The headphone wire acts as a $\sim$75 cm monopole antenna
- No external antenna is needed when headphones are plugged in

3.3.4 Mute Logic

Headphone insertion is detected by the PJ-320A jack's internal switch on D8:

1. Plug inserted → D8 state changes (HP_DETECT)
2. Firmware pulls D9 LOW
3. D9 drives PAM8403 $\overline{\text{SHDN}}$ LOW → amplifier enters shutdown
4. Speaker mutes; audio continues through headphones only

3.3.5 Volume Control

Volume is controlled digitally via the RDA5807M register 0x05, bits [3:0]. Range: 0 (mute) to 15 (maximum). No analogue potentiometer is used - the left rotary encoder adjusts this register in firmware.

3.4 USB-C Integration

A single USB-C receptacle serves both charging and programming - no separate power jack or FTDI cable required.

3.4.1 Pin Allocation

- **VBUS (5 V)** → TP4056 IN+ for battery charging
- **D+ / D-** → CH340C for USB-to-UART conversion
- **CC1 / CC2** → 5.1 kΩ pull-downs to GND (USB-C device identification per USB Type-C specification)

3.4.2 UART Bridge

The CH340C converts USB data to 3.3 V UART:

- CH340C TXD → ATmega328P RXD (PD0)
- CH340C RXD ← ATmega328P TXD (PD1)

3.4.3 Auto-Reset

The Arduino-standard auto-reset circuit allows the IDE to reset the MCU before upload:

CH340C DTR# → C14 (100 nF) → ATmega328P RESET

The capacitor differentiates the DTR edge into a brief reset pulse (∼1 ms) without holding RESET permanently low.

3.4.4 3.3 V Operation (V3 Pin)

The CH340C V3 (pin 4) **must** be tied to VCC for 3.3 V I/O level operation. A 100 nF bypass capacitor (C18) is placed between V3 and GND. This configures the internal voltage regulator to output 3.3 V on the data lines.

Power Switch Must Be ON for USB

The CH340C is powered from the **+3V3** rail, which is downstream of the latching power switch SW1. If SW1 is OFF, the CH340C has no power and the USB-C port appears completely dead to the host PC - no COM port enumerates. **The radio must be turned ON before attempting firmware upload.**

3.5 Complete Bill of Materials

Table 3 lists every component on the Phase 3 PCB.

Table 3: Phase 3 complete bill of materials.

Ref	Component	Value / Part	Pkg	Qty	Purpose
Integrated Circuits					
U1	Microcontroller	ATmega328P-PU	DIP-28	1	Main MCU (8 MHz, 3.3 V)
U2	FM Radio Module	RDA5807M (RRD-102)	Module	1	DSP FM receiver, I^2C
U3	Audio Amplifier	PAM8403	SOP-16	1	Class-D, mono BTL to speaker
U4	LiPo Charger	TP4056	SOP-8	1	CC/CV charger, 455 mA
U5	LDO Regulator	HT7333	SOT-89	1	3.3 V, <100 mV dropout
U6	USB-UART Bridge	CH340C	SOP-16	1	USB-to-serial, 3.3 V
U7	Battery Protect IC	DW01A	SOT-23	1	OVP / UVP / OCP control

Continued on next page...

Ref	Component	Value / Part	Pkg	Qty	Purpose
Q1	Dual N-FET	FS8205A	SOT-23-6	1	Low-side protection switches
	OLED Display	SSD1306 128×64	Module	1	0.96" I^2C display
Resistors					
R1, R2	Pull-up	4.7 kΩ	0805	2	I^2C bus (SDA, SCL)
R3, R4	Mixing	10 kΩ	0805	2	L+R mono summing network
R5	Pull-up	10 kΩ	0805	1	RESET pull-up
R6	Current limit	330 Ω	0805	1	Power LED
R7, R8	Divider	100 kΩ	0805	2	Battery voltage sensing
R9	PROG	2.2 kΩ	0805	1	TP4056 charge current (455 mA)
R10, R11	Pull-down	5.1 kΩ	0805	2	USB-C CC1/CC2
R12, R13	Current limit	1 kΩ	0805	2	CHRG/STDBY LEDs
R14	Sense	100 Ω	0805	1	DW01A current sense
Capacitors					
C1, C2	Bypass	100 nF	0805	2	ATmega328P VCC/AVCC
C3	Load	22 pF ×2	0805	2	Crystal load capacitors
C4	Bypass	100 nF	0805	1	RDA5807M supply
C5	Coupling	1 µF	0805	1	Audio DC block (mono path)
C6, C7	Coupling	10 µF	0805	2	Headphone DC block (L, R)
C8	Antenna	100 pF	0805	1	RF coupling to jack
C9, C10	Filter	10 µF	0805	2	HT7333 input/output
C11	Bypass	10 µF	0805	1	PAM8403 PVCC (ch. 1)
C12	Filter	10 µF	0805	1	TP4056 input
C13	Bypass	100 nF	0805	1	CH340C VCC
C14	Auto-reset	100 nF	0805	1	DTR# to RESET
C15	Filter	1 µF	0805	1	TP4056 output
C16	Bulk	10 µF	0805	1	RDA5807M bulk decoupling
C17	Filter	1 µF	0805	1	TP4056 BAT pin
C18	Bypass	100 nF	0805	1	CH340C V3 pin
C19	Filter	100 nF	0805	1	ADC input (A0)
C20	Bypass	10 µF	0805	1	PAM8403 PVCC (ch. 2)

Continued on next page...

Ref	Component	Value / Part	Pkg	Qty	Purpose
C21	Reference	100 nF	0805	1	ATmega AREF stability
Connectors & Mechanical					
J1	Audio Jack	PJ-320A	TH	1	3.5 mm stereo + HP detect
J2	Battery	JST PH 2-pin	TH	1	LiPo connection
J3	ISP Header	2×3 pin (2.54 mm)	TH	1	Bootloader / ISP programming
J4	USB-C	16-pin receptacle	SMD	1	Charging + data
J5	OLED Header	4-pin (2.54 mm)	TH	1	GND-VCC-SCL-SDA
J6, J7	Encoder Headers	5-pin (2.54 mm)	TH	2	Rotary encoder connection
J8	Radio Module	10-pin (1.27 mm)	TH	1	RDA5807M module socket
J9	Speaker	2-pin (2.54 mm)	TH	1	Speaker connection
SW1	Power Switch	Latching push	TH	1	Main power on/off
Y1	Crystal	8 MHz	HC-49S	1	MCU clock source

4 PCB Design & KiCad Workflow

This section documents the complete PCB design process - from tool selection and library creation through schematic entry, layout, and DRC resolution. The workflow reflects our realities as first-time KiCad users inheriting an incomplete project from a teammate.

4.1 Design Tool Selection

KiCad 9.0 was chosen as the EDA tool for the following reasons:

- **Free and open-source** - no licence cost or seat restrictions
- **University standard** - recommended by the department for coursework submissions
- **Full-featured** - schematic capture, PCB layout, 3D viewer, Gerber export, and DRC in a single package

4.2 Custom Symbol & Footprint Creation

Several components on the Phase 3 board have no standard KiCad library parts. Custom symbols and/or footprints were created for each:

- RDA5807M module - 11×11 mm, 2×5 pads, 1.27 mm pitch (castellation footprint, no standard library match)
- PAM8403 (SOP-16) - custom symbol with correct pin mapping
- TP4056 (SOP-8) - custom symbol

- CH340C (SOP-16) - custom symbol
- HT7333 (SOT-89) - custom symbol
- FS8205A (SOT-23-6) - custom symbol

LOG ENTRY 4 • Feb 17 • Custom RDA5807M symbol

First attempt at creating a custom KiCad symbol for the RDA5807M module. Made critical errors: used wrong pin types (all set to Passive instead of Bidirectional for SDA/SCL, Power I/O for VCC/GND, Output for LOUT/ROUT). Also had no pin numbers - only names. KiCad couldn't match symbol pins to footprint pads. Required complete rebuild of the symbol.

4.3 Library Path Issues

Library Path Hell

The KiCad project referenced the teammate's old PC paths (C:\Users\OtherUser\...) in fp-lib-table and sym-lib-table. Every footprint showed *"not found"*. Fix: replace all absolute paths with ${KIPRJMOD} variable for portability. Also discovered the TP4056 entry was pointing to the PAM8403 folder due to a copy-paste error.

4.4 Schematic Entry

After resolving library paths and rebuilding custom symbols, the schematic was entered and validated using KiCad's Electrical Rules Check (ERC).

4.4.1 ERC Results

- **Initial run:** 1 error + 18 warnings
- **Fixed:** power_pin_not_driven error - a power flag was missing on the battery net
- **Remaining warnings:** bidirectional pin conflicts (PAM8403 symbol pin types), unconnected pins (acceptable - unused MCU I/O)
- **Final ERC:** 0 errors, 4 warnings (SW1/SW2 mechanical pads - ignorable)

4.5 PCB Layout

4.5.1 Board Outline & Mechanical

- Board dimensions: 70 × 50 mm rectangle
- Corner treatment: 2 mm fillet (rounded corners)
- Mounting: M3 holes at all four corners

4.5.2 Design Rules

- Minimum trace width: 0.25 mm (signal traces)
- Power traces: 0.5–1.0 mm (VBAT, +3V3 distribution)
- Clearance: 0.2 mm minimum copper-to-copper

- Via drill: 0.3 mm

4.5.3 Layer Stackup

- **Top (F.Cu):** all components + signal traces
- **Bottom (B.Cu):** continuous GND copper pour (ground plane)

4.5.4 Zone Separation

The board is divided into functional zones to minimise interference:

- **Analog (west side):** `RDA5807M`, audio mixing network, headphone jack
- **Digital (east side):** `ATmega328P`, `SSD1306` OLED, encoders
- **Power (south-west):** `TP4056`, `DW01A`/`FS8205A`, `HT7333`, battery connector

4.5.5 Component Placement Rationale

- USB-C connector at bottom edge (accessible for cable insertion)
- OLED display at top (visible when radio is held)
- Rotary encoders at left and right sides (ergonomic thumb access)
- `RDA5807M` placed away from `PAM8403` (minimise Class-D switching noise coupling into RF front-end)

4.6 Autorouter Attempt

Autorouter Failure

First attempt used the Freerouting autorouter. Result: 137 DRC violations including shorts, no-net vias, and clearance violations. The autorouter created ghost traces and unconnected copper fragments. All autorouted traces were deleted and manual routing was performed instead.

4.7 Lessons from PCB Design

`${KIPRJMOD}` saves portability headaches
Always use project-relative paths in `fp-lib-table` and `sym-lib-table`. Absolute paths break the moment the project moves to another machine.

Custom symbols need correct pin types from the start
Setting all pins to `Passive` silences ERC but hides real wiring errors. Use `Power I/O`, `Bidirectional`, `Output`, etc. as appropriate.

Autorouters fail on mixed-signal boards
Manual routing is required for boards combining RF, analog audio, and digital logic. The autorouter cannot understand zone separation or sensitive signal isolation.

DRC markers can be hidden during routing
Appearance Panel → Objects → "DRC Markers" → click the eye icon. Markers still exist in the DRC list but won't clutter the canvas.

One ghost trace can generate dozens of DRC arrows
A single leftover copper fragment from incomplete routing creates a chain reaction of violations (shorts, clearance, unconnected nets). Always inspect and delete orphan traces.

Crystal must be within 10 mm of MCU oscillator pins
Long traces to the crystal add parasitic capacitance and can prevent oscillator startup. Keep Y1 as close to XTAL1/XTAL2 as physically possible.

Ground plane continuity requires stitching vias
Where components on the top layer block the bottom copper pour, add stitching vias on both sides of the gap to maintain a low-impedance return path.

4.8 Schematic & PCB Images

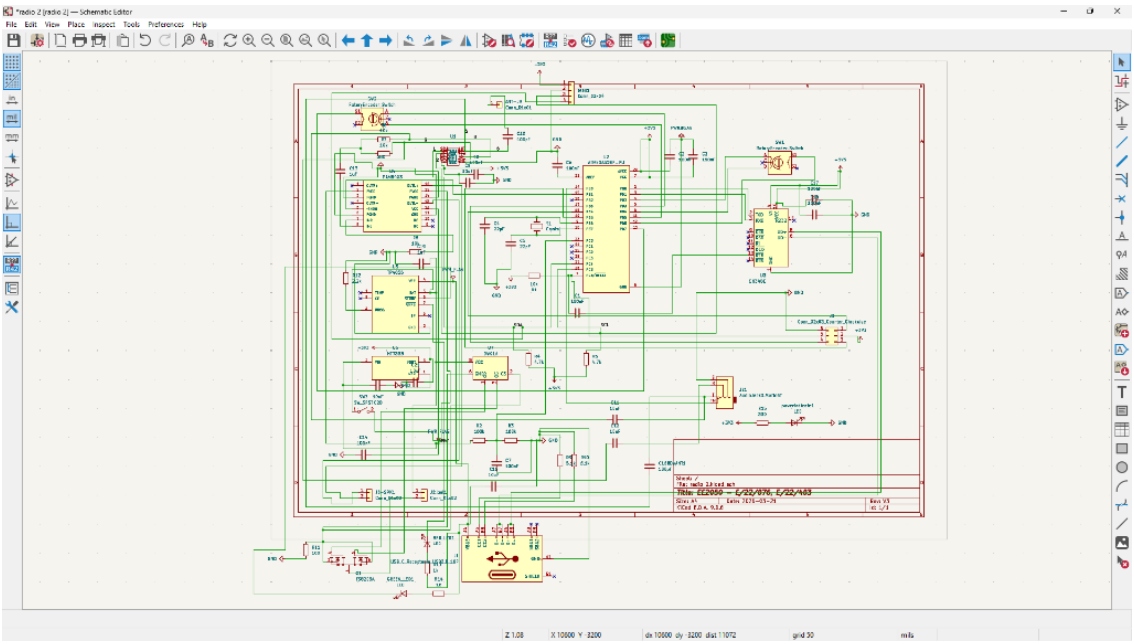


Figure 4: Phase 3 KiCad Schematic.

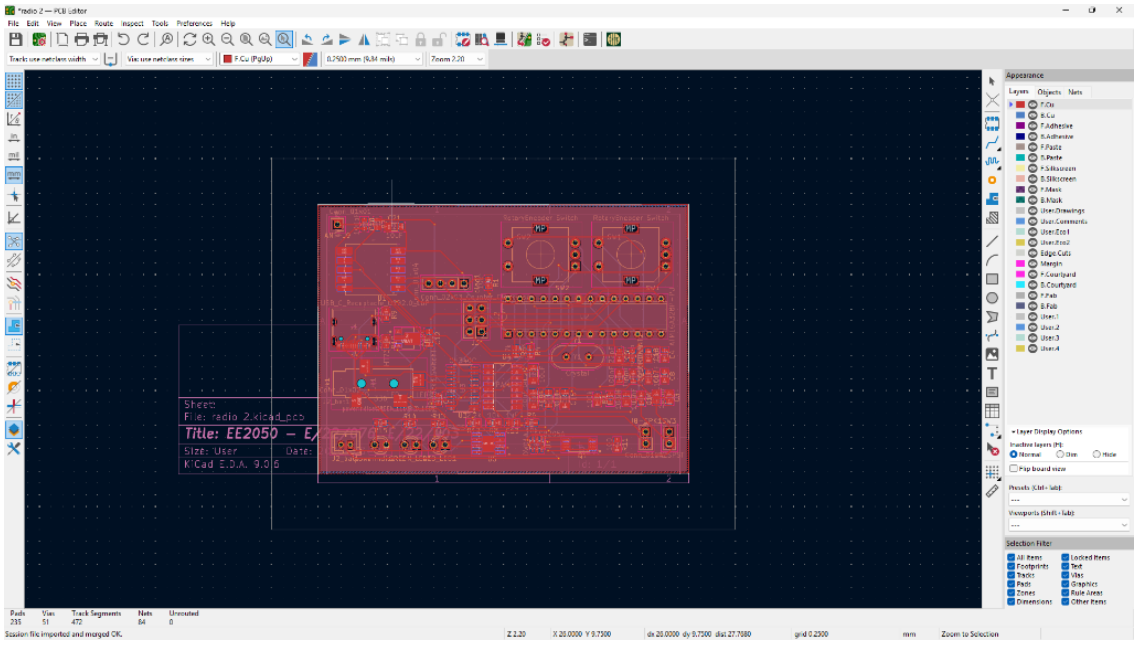


Figure 5: Phase 3 PCB layout (KiCad PCB Editor) showing fully routed traces.

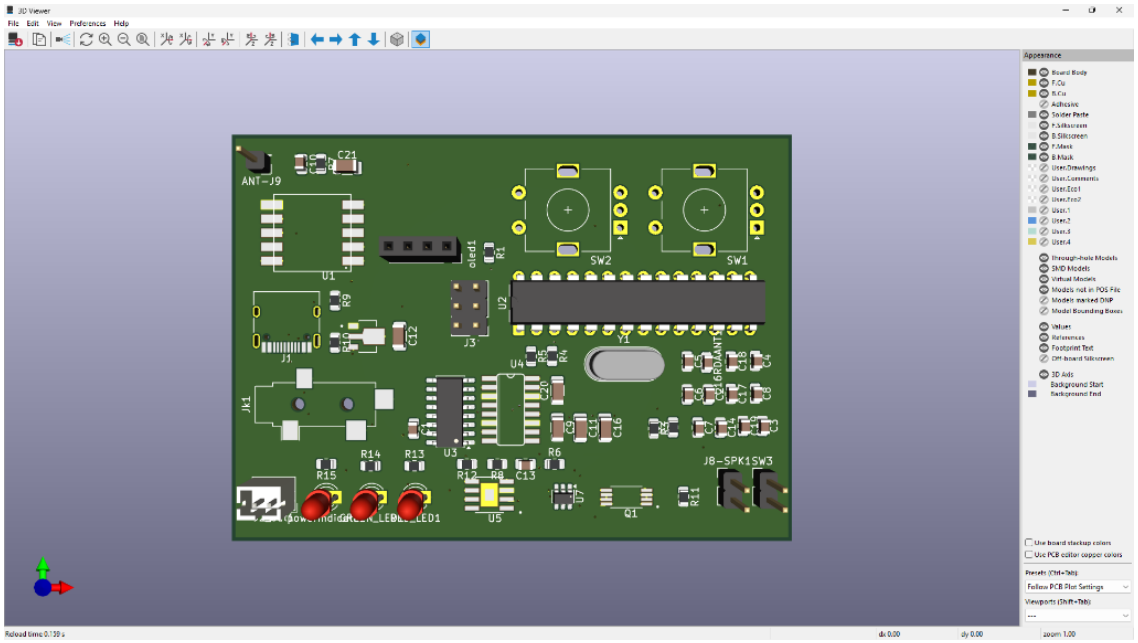


Figure 6: 3D render of the PCB (custom component models omitted).

5 Conclusion

This project achieved its primary objective - a working FM radio - in Iteration 1, and progressively enhanced it through two more iterations. The final design (Iteration 3) represents a product-grade architecture: custom PCB, integrated charging, single USB-C port, Class-D amplification, and OLED display with a 68-station database. The journey from a noisy perfboard to a complete PCB design demonstrates the iterative engineering process at its core.

The Phase 3 design was completed as a software deliverable ready for future fabrication. No design knowledge has been lost; every decision is traceable.

The AI-assisted methodology proved both powerful and dangerous. AI models accelerated the design process dramatically - generating build documents, debugging hardware issues in real-time, and assisting in comprehensive PCB verification and DRC review. But they also introduced errors (wrong IC compatibility claims, incorrect pinouts in early drafts) that required human verification against datasheets. The key lesson: AI is an excellent engineering partner, but never the final authority. The engineer must always remain the decision-maker.

6 References

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